

Sentinel

AI/NLP Risk Engine for Real-Time Market Risk

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Problem & Approach

THE PROBLEM

- ▶ **Risk desks drown in headlines.** Analysts can't read every wire, tweet, and filing intraday — material events get priced before they're read.
- ▶ **Sentiment alone isn't a signal.** A polarity score with no event type, no ticker linkage, and no impact estimate can't drive a portfolio decision.
- ▶ **Stress tests lag the news.** Scenario shocks are run quarterly on stale assumptions while the market reprices in minutes.

OUR APPROACH

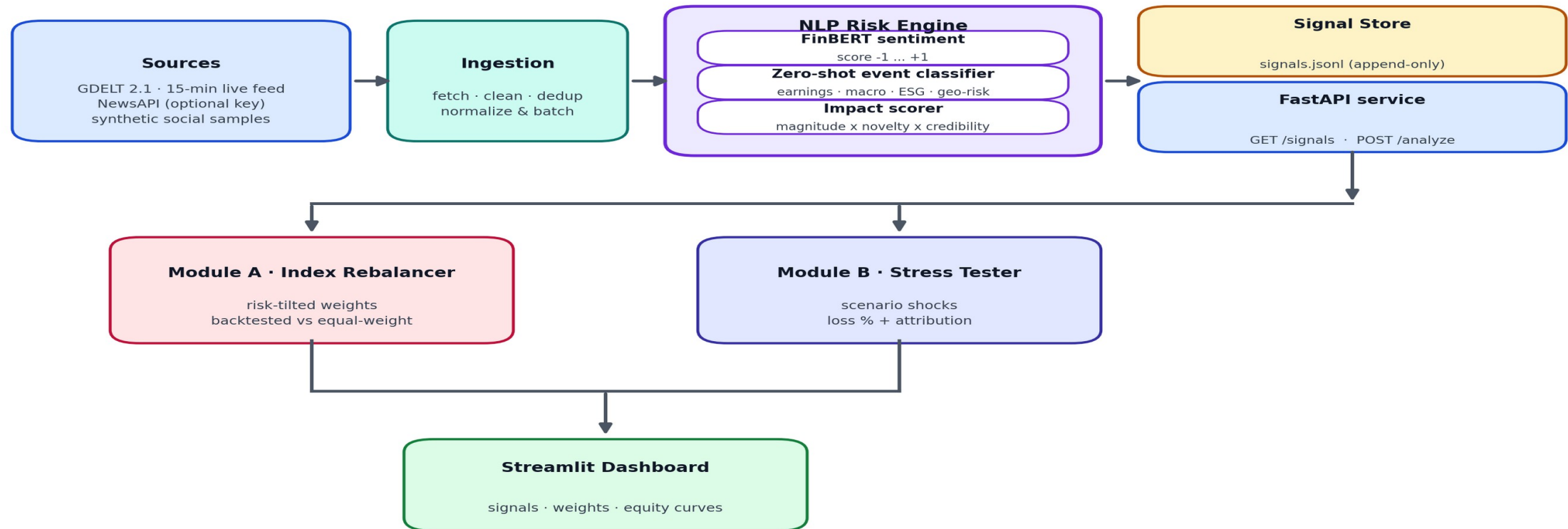
- ▶ **Three-stage NLP risk engine.** FinBERT-style sentiment → zero-shot event classification → impact scorer (magnitude × novelty × credibility).
- ▶ **Ticker-linked signal store.** Every signal lands in signals.jsonl and a FastAPI layer (/signals, /analyze) — one contract for all consumers.
- ▶ **Two downstream modules.** (A) Index rebalancer tilts weights away from negative-risk names; (B) stress tester prices event-driven scenarios with loss attribution.

System Design

Sources → Ingestion → NLP Risk Engine → Signal Store + API → Modules A/B → Dashboard

Sentinel — AI/NLP Risk Engine: Data Flow & System Architecture

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Runs fully on lightweight fallback backends — no GPU, no model downloads. Set `RISK_ENGINE_BACKEND=hf` (+ torch) for real FinBERT / zero-shot models.

Implementation Highlights

- ▶ **Modular pipeline (risk_engine / module_a / module_b / api / dashboard).** Each stage is runnable standalone — ``PYTHONPATH=src python -m risk_engine.pipeline --sources synthetic_social --limit 40``.
- ▶ **Zero-model default, real-model upgrade path.** VADER-style lexicon sentiment + heuristic event classifier run with no downloads; set `RISK_ENGINE_BACKEND=hf` with torch for FinBERT/zero-shot via transformers.
- ▶ **Keyless, offline-capable data.** GDELT 2.1 needs no API key; NewsAPI is optional; yfinance prices and the wholesale portfolio both have deterministic synthetic fallbacks.
- ▶ **Reproducible metrics pipeline.** Backtest and stress test write `data/module_a_results.json` / `data/module_b_results.json` — this deck is generated from those files.
- ▶ **Typed API + interactive dashboard.** FastAPI serves signals and ad-hoc analysis; Streamlit/Plotly renders weights, equity curves, and loss attribution.

Key Results

Module A backtest vs. equal-weight baseline · Module B event-driven stress test

MODULE A — INDEX REBALANCER

Signals processed

40

Sharpe — risk-tilted

2.665

Sharpe — baseline

2.676

Cum. return — tilted

17.93%

Cum. return — baseline

18.02%

Max drawdown

5.43%

Turnover

82.80%

Worst scenario loss

26.28%

Worst scenario

Geopolitical

Scenarios tested

3

Domain Impact



What a risk desk gains from Sentinel

- ▶ **News-to-action in minutes, not hours.** Ticker-linked risk signals flow intraday — no analyst required to read every headline first.
- ▶ **Systematic de-risking.** Module A trims exposure to names under negative event pressure before clusters compound.
- ▶ **Board-ready worst cases.** Module B turns 'what if rates spike?' into a loss % with per-position attribution a CRO can defend.
- ▶ **Zero vendor lock-in.** Keyless feeds, offline fallbacks, laptop-grade compute — deployable on a desk or a free-tier cloud box.
- ▶ **Auditable by construction.** Every signal, weight, and scenario result is a JSON artifact — reproducible and inspectable.

Limitations & Next Steps

LIMITATIONS

- ▶ Backtests ignore transaction costs and slippage.
- ▶ Entity→ticker mapping covers a fixed large-cap watchlist.
- ▶ Synthetic samples model headline structure, not market outcomes.
- ▶ Free-tier feeds can lag; no streaming WebSocket yet.

NEXT STEPS

- ▶ Fine-tune FinBERT on finance headlines; add multilingual ingestion.
- ▶ Live WebSocket push for breaking-signal alerts.
- ▶ Transaction-cost-aware rebalancing and turnover penalties.
- ▶ Expand scenario library (credit, liquidity, climate) with regulator templates.

Sentinel — real-time NLP risk signals, risk-tilted rebalancing, event-driven stress testing.